Appendix 2
Cremona Groups and Polynomial Rectification of Embedded Affine Lines: Results and a Research Programme

Ivan Dorofeev and Alexander Frolov
Researchers
qft98romanov@gmail.com

28 October 2026

Abstract

This research continuation studies the explicit rectifications of the Shastri trefoil and a harmonic figure-eight over $\mathbb{Q}(\sqrt{-3})$. The embedding problem is approached through the distinction between birational and polynomial equivalence, relative automorphisms, and the removal of poles. We present working proofs that the chosen rectifiers are wild, belong to the standard Cremona subgroup, and are stably tame. A family of transfers is constructed, and we distinguish the specific questions answered by the constructions, the inputs taken from the literature, and the questions that remain open. Symbolic certificates use exact arithmetic and are accompanied by reproducible code.

Acknowledgements. The authors express their sincere gratitude to Professor Alexei Kanel-Belov and Professor Alexander Gasnikov.

Status: a research branch; the entire appendix remains under verification. This branch continues the study of consequences of the constructed rectifiers and their connections with the Abhyankar–Moh embedding problem, Cremona groups, birational geometry, and affine geometry. Despite the preliminary status, the authors consider it useful to share these arguments and computations. The authors hope that they will develop into a full research programme contributing to the understanding of these areas, and invite all interested readers to check, discuss, correct, and extend the work.

The material was prepared in an AI-assisted research process. Internal reruns and alternative implementations do not constitute external independent peer review. The terms "theorem" and "proof" designate proposed mathematical arguments and do not override the overall verification status. No claim of publication priority is made. This appendix does not replace the original preprint or Appendix 1.

Source and scope of the update. The source repository is `qft98romanov-hash/abhyankar-moh-trefoil`, at commit `b954f3609e900761fddc9c8ff03d8301e9604cab`. The original formulas and previously released appendices are preserved. Here a question is called "settled" only in the sense of a logical consequence of the displayed formulas and cited theorems within this version; this does not assert a solution of a general problem or acceptance by the mathematical community. This English edition translates the Russian research version dated 28 October 2026 and retains its mathematical scope and caveats.

Contents

1 Established conclusions, excluded candidates, and open questions 3

2 Formulation and research programme 4

2.1 Lifting the parameter to a coordinate 5

2.2 What the two-dimensional case provides 5

2.3 Birational rectification and the precise group formulation 5

2.4 The relative subgroup as a working mechanism 7

2.5 The field of definition and the descent obstruction 7

2.6 What remains a programme rather than a result 8

3 Constructions used in the group-theoretic arguments 8

3.1 The coefficient field and the base coordinate pair 8

3.2 Transfer with control of the forward and inverse maps 9

3.3 The harmonic model and the family C_N 10

3.4 Integrality of the formulas 11

4 Wildness of the constructed rectifiers 11

4.1 The inner coordinate of a rational-length-two presentation 11

5 The standard Cremona subgroup 12

6 Stable tameness and the scope of the conclusion 13

7 Reproducibility and the limits of the computational certificate 14

1 Established conclusions, excluded candidates, and open questions

Status of this research branch. All statements in this appendix are presented as working proofs and computational certificates under verification. Internal checks of identities and repeated applications of published criteria are not external independent peer review. Below, the word *consequence* denotes a logical deduction from the displayed formulas and cited theorems; it does not assert priority in the literature.

Question	Conclusion and its precise scope
The specific Shastri curve	A polynomial rectifier over $K = \mathbb{Q}(\sqrt{-3})$ excludes this parametrization as a candidate for complex nonrectifiability. It does not exclude other nonrectifiable spatial lines.
The harmonic model and C_N	The specified parametrizations are rectified. The harmonic model uses $D_5 = t^5 - 5t^3 + 5t$; replacing $5t$ by $4t$ is not covered by this statement.
Tameness of the constructed maps	Wildness of Φ_T, Φ_H, Φ_N follows from relative-coordinate criteria. The nonexistence of *any* tame rectifier of the same curves has not been proved.
Stabilization	$\Phi_* \times \mathrm{id}_{\mathbb{A}^3} \in \mathrm{Tame}_6(K)$ follows from a known stable-tameness theorem. The minimum number of additional variables has not been determined.
Cremona subgroup	The constructed maps belong to $\mathcal{C}_3(K)$. This neither describes the whole group $\mathrm{Cr}_3(K)$ nor provides the first example of a wild element of $\mathcal{C}_3(K)$.
Field of definition	For Shastri's curve, the minimum degree of a suitable number field over $\mathbb{Q}$ is 2. Uniqueness of a suitable quadratic field and an effective classification of rectification fields are not established.

The historical candidate and the general problem. Shastri's paper [3, §1 and §3, Example 1] proposes real knot parametrizations as candidates for investigating complex nonrectifiability. Thus a complete polynomial automorphism for γ_T answers the rectification question for *this curve* affirmatively. It does not disprove the historical conjecture that at least one nonrectifiable embedding $\mathbb{A}^1_{\mathbb{C}} \hookrightarrow \mathbb{A}^3_{\mathbb{C}}$ exists. Nor does it prove the opposite universal conjecture that every such embedding is rectifiable. We do not attribute to Shastri's paper a separate numbered conjecture that it does not state.

Related conjectures from the first appendix. The consequences for the van den Essen–van Rossum candidates already recorded in the first appendix must be accounted for separately [6, §5, Conjectures 5.1–5.3].[1] These are consequences of the original rectification, rather than new results about the Cremona group. For $\gamma_T = (f, g, h)$, consider

$$D = t\partial_u + f'(u)\partial_x + g'(u)\partial_y + h'(u)\partial_z \quad \text{on } \mathbb{C}[t, u, x, y, z].$$

Conjecture 5.1, also called *Conjecture E* in subsequent literature, predicted that $\ker D$ is not a polynomial algebra. From the rectifier $\Phi_T = (P, Q, R)$ with inverse Ψ, setting $V = \gamma_T(u) - t(x, y, z)$ gives

$$p = P(V), \qquad q = Q(V)/t, \qquad r = R(V)/t, \qquad s = (u - P(V))/t.$$

[1] The numbering and identification of the objects here are inherited from Appendix 1. The publisher's record was checked in preparing this version, but a fresh download of the full AMS text was unavailable. The algebraic deductions below were checked directly; this limitation of the bibliographic audit is also recorded in `audit/claim_status.md`.

All divisions are exact because $\Phi_T\gamma_T = (u, 0, 0)$. The inverse change of variables is

$$u = p + ts, \qquad (x, y, z) = \frac{\gamma_T(p + ts) - \Psi(p, tq, tr)}{t};$$

it is polynomial as well. Consequently,

$$\mathbb{C}[t, u, x, y, z] = \mathbb{C}[t, p, q, r, s], \qquad D = \partial_s, \qquad \ker D = \mathbb{C}[t, p, q, r].$$

This negates precisely the specified conjecture 5.1. If σ denotes the original slice in the same construction, then $\sigma - s \in \ker D$, whence $\mathbb{C}[t, u, x, y, z]/(\sigma) \simeq \mathbb{C}^{[4]}$. Furthermore, for $I_{\gamma_T} = (Q, R)$ we have

$$\mathbb{C}[x, y, z][t, I_{\gamma_T}/t] = \mathbb{C}[t, P, Q/t, R/t] \simeq \mathbb{C}^{[4]}.$$

These are the corresponding polynomial descriptions of the objects in 5.2 and 5.3. Detailed formulas, identification of the notation, and certificates are provided in the repository's first appendix. These particular calculations do not solve the general cancellation, embedding, or action-linearization problems. Oda's withdrawn preprint [13] is not used as a proof: its arXiv record states that it was withdrawn on 9 May 2020 because of an error.

Predecessors of the programme. The connection between the embedding problem and Cremona equivalence is explicitly discussed by Mella–Polastri [8, introduction and Theorem 1]. Relative coordinates, factorizations over the fraction field, and transfer are studied by Edo–Vénéreau [5]. The closest predecessor of the constructive part is Lewis [12, Lemma 1, Theorems 2 and 12]: a coordinate restricting to the parameter is constructed using residual coordinates and corrections of rational transformations. That paper also compares earlier results of Craighero, Bhatwadekar–Roy, and Kuroda. Accordingly, the general procedure *localize, then eliminate denominators* is not claimed here as a new strategy.

The existence of wild automorphisms in three variables has long been established; the passage from a wild relative coordinate to wildness in three variables is supplied by [7]. Blanc–Hedén already proved Nagata $\in \mathcal{C}_3$ [11, Proposition 6.8]. Stable tameness for the class used here follows from [9, Corollary 4.9]. The contribution of this appendix consists in applying these results to the *specific* rectifiers, verifying their hypotheses, and formulating an explicit programme for further work.

Unresolved questions and limits of the search. Questions remaining open within this project include completeness of the relative method, the existence of a tame rectifier for the same curves, minimum rectifier degrees, descent to other totally imaginary number fields, and the general case $\mathbb{A}^1 \hookrightarrow \mathbb{A}^3$ over $\mathbb{C}$. Finite computations for $N \le N_{\max}$ check the implementation; statements for every N rely on inductive proofs. Failure of a bounded search does not prove nonrectifiability. A targeted literature search as of 28 October 2026 did not identify an earlier publication containing our complete Shastri rectifier; failure to find such a publication does not establish priority.

2 Formulation and research programme

Throughout this appendix, a *rectification* is an automorphism of the entire affine space that sends the given parametrized curve to the standard line. The choice of field and the requirement of a polynomial inverse are part of the problem. Birational rectification is a weaker condition.

2.1 Lifting the parameter to a coordinate

Let k be a field, let $R = k[x, y, z]$, and write

$$\gamma(t) = (p(t), q(t), r(t)), \qquad p, q, r \in k[t].$$

The parametrization defines a closed embedding $\mathbb{A}^1_k \hookrightarrow \mathbb{A}^3_k$ if and only if

$$k[p, q, r] = k[t]. \tag{1}$$

This is equivalent to surjectivity of the restriction homomorphism $\gamma^* : R \to k[t]$. In particular, there is a polynomial $H_0 \in R$ with $H_0\gamma = t$. Such a polynomial need not, however, be a *coordinate*, that is, a component of a polynomial automorphism.

Lemma 2.1 (Coordinate lift)**.** *The embedding γ admits a rectifier $\Phi \in \mathrm{Aut}_k(\mathbb{A}^3)$ with $\Phi\gamma(t) = (t, 0, 0)$ if and only if there is a coordinate $H \in R$ satisfying $H\gamma = t$.*

Proof. Necessity follows from the first component of Φ. Conversely, complete H to an automorphism (H, U, V). Its value on the curve is $(t, u(t), v(t))$. Follow it by the automorphism

$$(T, Y, Z) \longmapsto (T, Y - u(T), Z - v(T)).$$

The resulting composition rectifies γ. □

This criterion is known and is used explicitly in [12, Lemma 1]. Its constructive content is the choice of a coordinate among all lifts $H_0 + \ker \gamma^*$.

2.2 What the two-dimensional case provides

In characteristic zero, the Abhyankar–Moh–Suzuki theorem states that every closed embedding $\mathbb{A}^1_k \hookrightarrow \mathbb{A}^2_k$ is rectifiable [1, 2]. For unparametrized images, this means that the action of $G_2(k) = \mathrm{Aut}_k(\mathbb{A}^2)$ is transitive. Setting $L_2 = \{y = 0\}$ gives the following description of the set of such images:

$$\{\text{images of closed embeddings } \mathbb{A}^1_k \hookrightarrow \mathbb{A}^2_k\} \simeq G_2(k)/\operatorname{Stab}(L_2).$$

Here the quotient consists of left cosets $g \operatorname{Stab}(L_2)$, with

$$\operatorname{Stab}(L_2) = \{(x, y) \mapsto (ax + P(y), by) : a, b \in k^*,\ P \in k[y]\}. \tag{2}$$

Indeed, preservation of the ideal (y) forces the image of y to be by. After scaling, an automorphism of $k[y][x]$ remains, so the image of x is $ax + P(y)$ with $a \in k^*$. The reverse inclusion is immediate.

Three classical statements should be distinguished. The Abhyankar–Moh–Suzuki theorem describes the orbit of embedded lines. The Jung–van der Kulk theorem describes $G_2(k)$ through affine and triangular automorphisms and holds over every field [5, Theorem 1]. The Noether–Castelnuovo theorem states that $\mathrm{Cr}_2(\mathbb{C})$ is generated by projective automorphisms and the standard quadratic involution; see the discussion in [11, §1]. Describing the orbit of a line does not by itself describe the whole Cremona group. In positive characteristic, the general plane rectification theorem is false, so its conclusion cannot be transferred to an arbitrary field without qualification.

2.3 Birational rectification and the precise group formulation

Choosing the standard affine chart gives an inclusion

$$G = \mathrm{Aut}_k(\mathbb{A}^3) \subset B = \mathrm{Cr}_3(k) = \mathrm{Bir}_k(\mathbb{P}^3).$$

For $b \in B$, there is an exact criterion:

$$b \in G \quad \Longleftrightarrow \quad b^* R = R. \tag{3}$$

Indeed, this equality means that all components of both b and b^{-1} are polynomial. In terms of orders along irreducible polynomials $P \in R$, the condition is

$$v_P(b_i) \geq 0, \qquad v_P((b^{-1})_i) \geq 0 \quad \text{for every } P \text{ and } i.$$

This uses unique factorization in R. Poles along the hyperplane at infinity are allowed. Checking the Jacobian alone does not replace condition (3).

Proposition 2.2. *Over any field k, every embedding (1) has a birational rectifier over k. The rectifier and its inverse can be chosen regular near the generic points of the respective curves. The rational Jacobian determinant of the rectifier can be made equal to 1.*

Proof. Differentiating $H_0(p, q, r) = t$ gives

$$(H_0)_x(p, q, r)p' + (H_0)_y(p, q, r)q' + (H_0)_z(p, q, r)r' = 1.$$

Thus, after cyclically permuting the coordinates, we may assume $p' \neq 0$. The extension $k(t)/k(p)$ is finite and separable, and $k(t) = k(p)(q, r)$. By the primitive element theorem, we can choose $\lambda(p) \in k[p]$ such that $k(p, q + \lambda(p)r) = k(t)$. Indeed, only finitely many coefficients $\lambda \in k(p)$ are bad, whereas $k[p]$ is infinite even when k is finite. This also follows directly by comparing the distinct embeddings of the separable extension: each pair of distinct embeddings excludes at most one value of λ.

Apply the shear $(x, y, z) \mapsto (x, y + \lambda(x)z, z)$ and again denote the second component of the curve by q. There is a $\tau \in k(x, y)$ satisfying $\tau(p, q) = t$; the denominator of the chosen expression does not vanish identically on the curve. Set

$$s = z - r(\tau(x, y)) + \tau(x, y), \qquad b = (s, x - p(s), y - q(s)). \tag{4}$$

Then $b\gamma = (t, 0, 0)$. For input (T, U, V), the inverse is given by

$$\begin{aligned} X = U + p(T), \qquad Y = V + q(T), \\ b^{-1}(T, U, V) = (X, Y, T + r(\tau(X, Y)) - \tau(X, Y)). \end{aligned} \tag{5}$$

Both compositions are the identity by direct substitution. On the line $(T, 0, 0)$, the denominator becomes the same nonzero element of $k(T)$, so the inverse is also defined at the generic point. All shears used have Jacobian determinant 1, as do the cyclic permutations. □

Over an algebraically closed field, birational equivalence of the projective closure of our curve to a line is a special case of the Mella–Polastri theorem: two birational embeddings of a d-dimensional variety into $\mathbb{P}^n$ are Cremona equivalent when $n \geq d + 2$ [8, Theorem 1]. The elementary proof above specifies a map over the original field. Neither statement guarantees regularity at every point of affine space.

For Shastri's curve, one can explicitly take

$$\tau(x, y) = \frac{x(y + 1)}{x^2 - y - 3}, \qquad s = z - \tau^5 + 11\tau, \qquad b = (s, x - s^3 + 3s, y - s^4 + 4s^2).$$

The identity $p(q + 1) = t(p^2 - q - 3)$, for $p = t^3 - 3t$ and $q = t^4 - 4t^2$, verifies recovery of the parameter. This is a birational rectifier over $\mathbb{Q}$ with poles allowed; both compositions with its explicit inverse are checked in the new computational package. Its existence is compatible with the impossibility of polynomial rectification over $\mathbb{Q}$ discussed below.

Let $L = \{y = z = 0\}$. Denote by S_L the subgroup of B consisting of birational maps that induce isomorphisms between some neighbourhoods of the generic point of L and act identically on L. Thus the regularity requirement applies to the inverse map as well.

Proposition 2.3 (The coset-intersection problem)**.** *If b is a birational rectifier from Proposition 2.2, then*

$$\gamma \text{ is polynomially rectifiable} \iff S_L b \cap G \neq \varnothing. \tag{6}$$

Proof. For a polynomial rectifier Φ, the map $h = \Phi b^{-1}$ belongs to S_L, and $\Phi = hb$. Conversely, if $hb \in G$ with $h \in S_L$, then $(hb)\gamma(t) = (t, 0, 0)$ as a rational identity. Since both sides are polynomial, the identity holds everywhere. □

The original problem is therefore reformulated as the search for a correction h that removes the affine poles of b and its inverse while preserving the parametrized line. Formula (6) is not by itself an algorithm and does not prove that the required intersection is nonempty.

2.4 The relative subgroup as a working mechanism

The stabilizer of the rational function x and its polynomial part are

$$J_x = \{b \in B : b^* x = x\} \simeq \mathrm{Cr}_2(k(x)), \qquad G_x = G \cap J_x = \mathrm{Aut}_{k[x]}\, k[x, y, z].$$

An element of G_x becomes a tame automorphism in two variables over $k(x)$, but its factorization over that field may have denominators. Eliminating them over $k[x]$ is the substantive part of the construction. This approach has direct predecessors in the theory of coordinate transfer and residual coordinates [5, 12].

If $A = (x, H, W) \in G_x$, $H\gamma = t$, and $W\gamma = w(t)$, then

$$\Phi = (H, x - p(H), W - w(H)) \tag{7}$$

rectifies γ. Within G_x itself, the natural target parametrization is $(p(t), t, 0)$: maps fixing x do not change the first component $p(t)$. A subsequent external tame automorphism sends this parametrization to $(t, 0, 0)$.

Remark 2.4. For a fixed choice of x, the existence of a relative coordinate H with $H\gamma = t$ is sufficient for rectifiability. Its necessity for an arbitrary rectifiable embedding is not proved here. Allowing an arbitrary preliminary polynomial change of variables without further restrictions would make a completeness claim circular: that change could itself be the rectifier being sought.

2.5 The field of definition and the descent obstruction

For Shastri's curve

$$\gamma_T = (t^3 - 3t, t^4 - 4t^2, t^5 - 10t),$$

the real image represents a trefoil [3, §3, Example 1]. A polynomial automorphism of $\mathbb{R}^3$ is a proper diffeomorphism and extends to a homeomorphism of the one-point compactification S^3. It cannot send a nontrivial knot to an unknot. Therefore rectification over $\mathbb{R}$, and hence over $\mathbb{Q}$, is impossible. The same argument applies to any number field with a real embedding: apply that embedding to the coefficients of a hypothetical rectifier.

Together with the explicit rectifier over $K = \mathbb{Q}(\sqrt{-3})$, this yields

$$\min\{[F : \mathbb{Q}] : F \text{ a number field, } \gamma_T \text{ rectifiable over } F\} = 2.$$

A suitable number field must be totally imaginary. Uniqueness of K among quadratic fields is not established; the absence of a root of $a^2 - a + 1$ in a field excludes direct specialization of our formula, but not a different construction.

Polynomial descent has a precise group formulation. Let $\alpha \in \mathrm{Aut}_K(\mathbb{A}^3)$ rectify γ_T, let $\Gamma = \mathrm{Gal}(K/\mathbb{Q})$, and let $H_L(K)$ be the pointwise *polynomial* stabilizer of L. Then

$$c_\sigma = \alpha\,\sigma(\alpha)^{-1} \in H_L(K), \qquad c_{\sigma\tau} = c_\sigma\,\sigma(c_\tau).$$

A rectifier over $\mathbb{Q}$ exists if and only if there is an $h \in H_L(K)$ with $c_\sigma = h^{-1}\sigma(h)$ for every σ. Indeed, this is equivalent to $h\alpha$ and its polynomial inverse being defined over $\mathbb{Q}$. For Shastri's curve, no such h exists. If the polynomial stabilizer is replaced by S_L, the obstruction disappears: for a birational rectifier b over $\mathbb{Q}$, one can take $h = b\alpha^{-1}$. Passing to the full Cremona group thus loses, among other things, information about polynomial descent. Only an elementary cocycle identity is used here; no classification of the nonabelian set $H^1(\Gamma, H_L(K))$ is asserted.

2.6 What remains a programme rather than a result

1. **Completeness of transfers.** Describe a class of parametrizations for which the choice of modification centres and exceptional divisors always leads to a relative coordinate. Finite computations for individual indices do not establish such completeness.
2. **Controlled changes of projection.** Determine which prescribed affine or bounded polynomial changes extend the applicability of G_x without a circular formulation.
3. **The tame orbit of the curve itself.** Wildness of the constructed rectifier does not exclude another tame rectifier. All coordinates in $H + \ker\gamma^*$ must be studied, rather than only the chosen H.
4. **Arithmetic.** Determine other fields of rectification; for example, test whether Shastri's curve has a rectifier over $\mathbb{Q}(i)$. The absence of a root needed by a particular formula and an obstruction to the existence of every automorphism are different statements.

These aims do not require a complete description of $\mathrm{Cr}_3(\mathbb{C})$ in advance. They concern specific subgroups, cosets, and regularity conditions on a fixed affine model.

3 Constructions used in the group-theoretic arguments

3.1 The coefficient field and the base coordinate pair

Set

$$K = \mathbb{Q}[a]/(a^2 - a + 1), \quad b = 1 - a, \quad ab = 1, \quad b^2 = -a.$$

The integrally normalized Chebyshev polynomials are defined recursively by $D_0 = 2$, $D_1 = t$, $D_{n+1} = tD_n - D_{n-1}$. In characteristic zero, $D_n(t) = 2T_n(t/2)$. In particular,

$$D_3 = t^3 - 3t, \quad D_4 = t^4 - 4t^2 + 2, \quad D_5 = t^5 - 5t^3 + 5t, \quad D_7 = t^7 - 7t^5 + 14t^3 - 7t.$$

We consider the specific parametrizations

$$\gamma_T = (D_3, D_4 - 2, t^5 - 10t), \qquad \gamma_H = (D_3, D_5, D_7).$$

The coefficient of t in the second component of γ_H is 5. The parametrization with coefficient 4 is not covered by this result.

The affine map $M = (x, y+2, z-5x)$ sends γ_T to $\gamma_0 = (D_3, D_4, D_5)$. In the normalized coordinates, introduce

$$\begin{aligned} u &= axy + z, & v &= -y, & m &= bx^5, \qquad s = bx, \\ \ell &= -1 + ax^2 - ax^4, & g &= -2x^5 + (6-2a)x^3 - 2bx, & & \\ P_x(U) &= sU^2 + \ell U + g, & H &= P_x(u) + mv. & & \end{aligned}$$

The polynomial P_x induces an automorphism of $K[x,T]/(x^5)$. We use the following representative of its inverse:

$$\begin{aligned} F_x(T) = & -T + x(bT^2 + 2a - 2) + x^2(2aT^3 - 5aT) \\ & + x^3(-5T^4 + 15T^2 - 2a) \\ & + x^4\big(14bT^5 + 50(a-1)T^3 + (29 - 24a)T\big). \end{aligned} \tag{8}$$

Direct computation gives both congruences

$$P_x(F_x(T)) \equiv T \pmod{x^5}, \qquad F_x(P_x(U)) \equiv U \pmod{x^5}.$$

Consequently,

$$W = \frac{u - F_x(H)}{m}, \qquad \chi_x(T) = \frac{T - P_x(F_x(T))}{m}$$

are polynomials. For $A_0 = (x, H, W)$, the inverse map is given by

$$\begin{aligned} U &= F_x(T) + mV, \\ V_0 &= \chi_x(T) - (\ell + 2sF_x(T))V - smV^2, \\ A_0^{-1}(x,T,V) &= (x, -V_0, U + axV_0). \end{aligned} \tag{9}$$

Indeed, expanding the square gives $P_x(U) + mV_0 = T$ and $(U - F_x(T))/m = V$. For the composition in the other direction, one first recovers u, v and then y, z. This is a polynomial proof of invertibility; constancy of the Jacobian alone is not used as a criterion.

Substitution into the curve gives

$$H(D_3, D_4, D_5) = t, \qquad W(D_3, D_4, D_5) = 8 - 14t^2.$$

Thus, with $h_T = H \circ M$ and $w_T = W \circ M$, we obtain

$$\Phi_T = (h_T,\ x - h_T^3 + 3h_T,\ w_T - 8 + 14h_T^2), \qquad \Phi_T\gamma_T = (t, 0, 0). \tag{10}$$

Its inverse is constructed from (9), M^{-1} and the inverses of the outer shears. The component degrees of this map are $(5, 15, 24)$, and its Jacobian is 1. No optimality of these degrees is claimed. The original implementation is in the repository's `verify.py`; the new verification code separately reproduces the compact certificates.

3.2 Transfer with control of the forward and inverse maps

The following mechanism is related to residual coordinates [4, 5, 12]; we record the formulas needed for reproducibility.

Proposition 3.1 (A sufficient transfer criterion)**.** *Let $A = (x, H, W)$ and $A^{-1} = (x, F, G)$ be mutually inverse polynomial maps over k. Let $0 \neq m \in k[x]$, $p \in k[x, y]$, and*

$$\beta = (x, p(x, y) + m(x)z, y).$$

If $c(y) = \overline{H(x, p(x, y), y)}$ defines an automorphism of $(k[x]/(m))[y]$, then $H \circ \beta$ extends to a relative polynomial automorphism.

Proof. Let ψ be the inverse of c in $B[T]$, where $B = k[x]/(m)$. Choose a lift of the polynomial

$$\varphi(x, T) \equiv W(x, p(x, \psi(T)), \psi(T)) \pmod{m}.$$

Then set

$$A_{\mathrm{new}} = \left(x, H \circ \beta, \frac{W \circ \beta - \varphi(x, H \circ \beta)}{m}\right). \tag{11}$$

The numerator of the last component vanishes modulo m. For input (x, T, V), put $S = mV + \varphi(x, T)$, $Y = F(x, T, S)$, $Z = G(x, T, S)$. The inverse map is

$$A_{\mathrm{new}}^{-1}(x, T, V) = \left(x, Z, \frac{Y - p(x, Z)}{m}\right). \tag{12}$$

In $B[T]$, invertibility of A yields $Y = p(x, \psi(T))$, $Z = \psi(T)$, so the division is again exact. After inverting m, these formulas are compositions of explicitly invertible maps. Injectivity of localization carries both identity compositions back to $k[x, y, z]$, including all fibres with $m = 0$. □

The condition on $c = c_0 + c_1 y + \sum_{j \geq 2} c_j y^j$ is equivalent to c_1 being a unit and all c_j, $j \geq 2$, being nilpotent in B. If $J = (c_j : j \geq 2)$ and $J^e = 0$, the inverse is obtained by the finite iteration

$$\psi_0 = c_1^{-1}(T - c_0), \qquad \psi_{i+1} = c_1^{-1}\left(T - c_0 - \sum_{j \geq 2} c_j \psi_i^j\right).$$

Differences between successive iterates lie in successively higher powers of J; after e steps a fixed point is reached. Thus the computations must be performed modulo the full m, rather than only its radical.

3.3 The harmonic model and the family C_N

The identity

$$D_3 D_7 - D_5^2 + 2 = D_4$$

shows that $\beta_1 = (x, xz - y^2 + 2, y)$ sends γ_H to γ_0. Since $H(0, Y, Z) = -Z$, the transfer centre is $-y$ modulo x, and Proposition 3.1 applies. One may take the correction

$$\varphi_1(T) = 42(a-1)T^6 + 175(1-a)T^4 + (147a - 161)T^2 + 8.$$

Putting $H_1 = H \circ \beta_1$ and $W_1 = (W \circ \beta_1 - \varphi_1(H_1))/x$, we have

$$W_1 \gamma_H = b(42t^3 - 49t), \quad \Phi_H = (H_1, x - H_1^3 + 3H_1, W_1 - b(42H_1^3 - 49H_1)).$$

The component degrees are $(7, 21, 41)$. The inverse is given by (12) and the inverse outer shears.

Now let $N \geq 1$. In $\mathbb{Z}[x, y]/(x^N)$, define

$$\nu_0 = -y, \qquad \nu_{i+1} = -y + x(\nu_i^2 - 2), \quad 0 \leq i < N, \qquad p_N \equiv 2 + x\nu_N - \nu_N^2 \pmod{x^N},$$

choosing the remainder p_N of degree less than N in x. We have $p_N \equiv 2 - y^2 \pmod{x}$. Set

$$C_N(t) = \left(D_3, D_5, \frac{D_4 - p_N(D_3, D_5)}{D_3^N}\right), \qquad \beta_N = (x, p_N + x^N z, y). \tag{13}$$

The division in this parametrization is exact. Indeed, $D_5 = D_3(t^2 - 2) - t$ and $D_4 = D_3 t - t^2 + 2$. At each step, the iteration error $\nu_i(D_3, D_5) - t$ acquires an additional factor D_3. Hence $\nu_N(D_3, D_5) \equiv t \pmod{D_3^N}$, which gives the required divisibility.

The centre $H(x, p_N, y)$ equals $-y$ modulo x. Therefore, modulo x^N, its linear coefficient is a unit and all higher coefficients are nilpotent. Proposition 3.1 constructs $A_N = (x, H_N, W_N)$ with $H_N = H \circ \beta_N$ and $H_N C_N = t$. With $w_N = W_N C_N$, we obtain

$$\Phi_N = (H_N, x - D_3(H_N), W_N - w_N(H_N)), \qquad \Phi_N C_N = (t, 0, 0). \tag{14}$$

Thus C_N does indeed define a closed embedding. For $N = 1$, one may choose the transfer above and obtain $C_1 = \gamma_H$, $\Phi_1 = \Phi_H$. Different representatives of the correction may change the companion by a tame shear; all subsequent statements concern maps constructed in this way.

3.4 Integrality of the formulas

All base coefficients lie in $\mathcal{O} = \mathbb{Z}[a]/(a^2 - a + 1)$. The congruences for P_x, F_x already hold over $\mathcal{O}$, and $b^{-1} = a$. Division by bx^5 consists of multiplication by a and exact division by the monomial x^5. The recursions for p_N and for the inverse of the centre introduce no numerical denominators: the linear coefficient of the centre is -1 modulo x and is inverted by a finite geometric series modulo x^N. Consequently, the polynomial maps and their inverses specialize to any field containing a root of $a^2 - a + 1$, including characteristics 2 and 3. This is a statement about the formulas; the wildness proofs below apply in characteristic zero. Degrees may decrease after specialization. For $\mathbb{F}_q$, a root exists if and only if $\operatorname{char} \mathbb{F}_q = 3$ or $q \equiv 1 \pmod 3$: in characteristic different from 3, the element $-a$ is a nontrivial cube root of unity. The absence of a root does not rule out other rectifiers.

4 Wildness of the constructed rectifiers

Throughout this section, $K = \mathbb{Q}[a]/(a^2 - a + 1)$, $b = 1 - a$, and $R = K[x]$. We use the identities $ab = 1$ and $b^2 = -a$. The tame group $\mathrm{Tame}_3(K)$ is generated by affine and elementary polynomial automorphisms. A coordinate is called tame if it is a component of a tame automorphism, and wild otherwise. Relative tameness over R means tameness in two variables over the ring R; a priori, this is stronger than tameness over the coefficient field K. The deductions from published theorems below are working proofs within this research branch and are submitted for further verification.

Proposition 4.1 (The Shastri rectifier)**.** *The coordinate h_T and the constructed rectifier Φ_T are wild over K.*

Proof. In the relative coordinates $(u, v) = (axy + z, -y)$, the base coordinate is $H = mv + P_x(u)$, where $m = bx^5$, and the coefficient of u^2 is bx. The Edo–Vénéreau criterion for length-one coordinates [5, Proposition 2] requires every nonlinear coefficient to be divisible by m for tameness. However, $bx^5 \nmid bx$ in R. The change $(y, z) \mapsto (u, v)$ is tame over R, so H is wild over R.

By the Umirbaev–Yu theorem, a wild $K[x]$-coordinate remains a wild coordinate of $K[x, y, z]$ in characteristic zero [7, Main Theorem]. The preliminary change $M = (x, y + 2, z - 5x)$ is tame, so $h_T = H \circ M$ is also wild. A tame automorphism cannot have a wild component; therefore Φ_T is wild. □

4.1 The inner coordinate of a rational-length-two presentation

For the next lemma, Y, Z denote two variables over R. We call a presentation

$$F = p_2 Y + Q_2(p_1 Z + Q_1(Y))$$

reduced in the sense of Edo if p_1 is monic, $Q_1 \in R[Y]$, $Q_1(0) = Q_2(0) = 0$, and $\gcd(p_1, \text{coefficients of } Q_1) = 1$; here $p_1, p_2 \neq 0$, $p_2 \in K(x)$, and $Q_2 \in K(x)[Y]$. We use the case $\deg Q_1 \geq 2$, $\deg Q_2 \geq 2$.

Lemma 4.2 (A consequence of Edo's classification)**.** *Suppose that $F \in R[Y, Z]$ is a coordinate with the reduced presentation above. If F is tame over R, then $p_1 Z + Q_1(Y)$ is also tame over R.*

Proof. Here products $\sigma\tau$ are compositions of R-algebra automorphisms: $(\sigma\tau)(f) = \sigma(\tau(f))$. Edo's Theorem 8(1) gives $F = \sigma\tau(Y)$, where both components of each of σ, τ have length one; such automorphisms are tame. As in the proof of Theorem 8(3), normalize $\sigma(Y) = p_3 Z + Q_3(Y)$,

where p_3 is monic and $Q_3(0) = 0$. This normalization is affine and preserves tameness. The Russell–Sathaye companion formula has the form

$$\sigma(Z) = \frac{\varepsilon}{p_3}(V(p_3 Z + Q_3(Y)) - Y), \qquad \varepsilon \in R^\times, \quad V \in R[Y].$$

If $\tau(Y) = p_4 Z + Q_4(Y)$, then

$$F = -\frac{\varepsilon p_4}{p_3} Y + \left(\frac{\varepsilon p_4}{p_3} V + Q_4\right)(p_3 Z + Q_3(Y)).$$

After removing the constant term, this is a reduced presentation of the same polynomial. Its uniqueness, established in Edo's Proposition 2, gives $p_3 = p_1$, $Q_3 = Q_1$. Thus the inner polynomial is a component of the tame automorphism σ; see also the same normalization in the proof of Theorem 8(3) [10]. □

Proposition 4.3 (The harmonic model and all constructed C_N)**.** *For every $N \geq 1$, the coordinate H_N and the constructed rectifier Φ_N are wild over K. In particular, this holds for the harmonic model $C_1 = \gamma_H$.*

Proof. Set

$$Q_x(U) = bxU^2 + (-1 + ax^2)U + g(x).$$

From $P_x(U) = Q_x(U) - ax^4 U$ and $a^2 + b = 0$, we obtain the identity

$$H_N = H(x, p_N + x^N z, y) = -ax^4 y + Q_x(ax^{N+1} z + axp_N + y).$$

Introduce

$$G_N = x^{N+1} z + by + x\big(p_N(x, y) - p_N(x, 0)\big),$$
$$c_N = axp_N(x, 0).$$

Then

$$H_N - Q_x(c_N) = -ax^4 y + [Q_x(aG_N + c_N) - Q_x(c_N)]. \tag{15}$$

This is a reduced presentation as in Lemma 4.2, with $p_1 = x^{N+1}$, $p_2 = -ax^4$, and $Q_1(y) = by + x(p_N(x, y) - p_N(x, 0))$. The linear coefficient of Q_1 is b modulo x, so the coprimality condition holds. The outer polynomial has degree two, and $\deg_y Q_1 \geq 2$ because $p_N \equiv 2 - y^2 \pmod{x}$.

Modulo x^{N+1}, the linear coefficient of Q_1 is invertible and all coefficients of y^j, $j \geq 2$, are nilpotent. The Russell–Sathaye criterion shows that G_N is an R-coordinate. However,

$$[y^2]Q_1 = -x \pmod{x^2}, \qquad x^{N+1} \nmid [y^2]Q_1 \quad (N \geq 1).$$

By the length-one criterion, G_N is wild over R. Lemma 4.2 rules out tameness of $H_N - Q_x(c_N)$, and subtracting a polynomial in x does not change tameness. The Umirbaev–Yu theorem gives wildness of H_N in three variables. Since H_N is a component of Φ_N, the latter is also wild. □

This proof applies to every N; it is not inferred from testing finitely many indices. For $N = 1$, its inner polynomial is $G_1 = x^2 z + by - xy^2$. Computational checks of the normalization and coefficients verify the calculations but do not replace Lemma 4.2.

5 The standard Cremona subgroup

Fix the notation

$$\mathcal{C}_3(K) = \langle \mathrm{PGL}_4(K), \sigma_3 \rangle, \qquad \sigma_3[X_0 : X_1 : X_2 : X_3] = [X_1X_2X_3 : X_0X_2X_3 : X_0X_1X_3 : X_0X_1X_2].$$

In this section, products of maps act from right to left. By Blanc–Hedén, $\mathrm{Tame}_3(K) \subset \mathcal{C}_3(K)$. Their monomial criterion in dimension three states that a monomial birational map with exponent matrix M belongs to $\mathcal{C}_3(K)$ if and only if every column sum of M is odd [11, Theorem 1.1 and Proposition 6.5].

Proposition 5.1. *All constructed rectifiers Φ_T, Φ_H, Φ_N belong to $\mathcal{C}_3(K)$.*

Proof. Consider the subgroup $\mathcal{L} \subset \mathrm{Cr}_3(K)$ fixing x and generated by

$$E_y(q) = (x, y + q(x,z), z), \quad E_z(q) = (x, y, z + q(x,y)),$$
$$S = (x, z, y), \qquad B_r = (x, x^r y, x^{-r} z),$$

where $q \in K[x, x^{-1}, T]$, $r \in \mathbb{Z}$, together with the constant diagonal maps (x, cy, dz), $c, d \in K^\times$.

All generators belong to $\mathcal{C}_3(K)$. For B_r, the column sums of the exponent matrix are $(1,1,1)$. For $D = (x, x^{2s}y, z)$, they are $(1+2s, 1, 1)$. Choose $s \geq 0$ so that $x^{2s}q$ is a polynomial. Then

$$E_y(q) = D^{-1} E_y(x^{2s} q) D.$$

The middle map is tame. The maps $E_z(q)$ are obtained by conjugation with S; the remaining constant generators are linear.

In the variables (u, v), with x fixed, the base map factors as

$$(u,v) \longmapsto \left(P_x(u) + mv, \frac{u - F_x(P_x(u) + mv)}{m} \right) = T_F D_m S T_{P/m},$$

where

$$T_{P/m}(u,v) = (u, v + P_x(u)/m), \quad D_m(u,v) = (mu, v/m),$$
$$T_F(u,v) = (u, v - F_x(u)/m), \qquad S(u,v) = (v,u).$$

For $m = bx^5$, every factor belongs to $\mathcal{L}$. The initial change $(y,z) \mapsto (axy + z, -y)$ is also a product of generators of $\mathcal{L}$. Hence $A_0 \in \mathcal{L}$.

For the transfer, set $D_N = (x, y, x^N z)$ and use the correction polynomial $\varphi_N(x,T)$ from the construction of the full companion. Then

$$\beta_N = S E_z(p_N) D_N, \qquad A_N = D_N^{-1} (E_z(-\varphi_N) A_0 S E_z(p_N)) D_N.$$

The map D_N normalizes $\mathcal{L}$, since

$$D_N^{-1} E_y(q) D_N = E_y(q(x, x^N T)),$$
$$D_N^{-1} E_z(q) D_N = E_z(x^{-N} q),$$
$$D_N^{-1} S D_N = B_N S,$$

and it commutes with B_r and the constant diagonal maps. The same formulas with $-N$ give the reverse inclusion. Therefore $A_N \in \mathcal{L} \subset \mathcal{C}_3(K)$. The outer maps completing the rectifications are tame, which proves the assertion. □

Remark 5.2. For odd N, the map D_N itself fails the monomial criterion: its first column sum is $1 + N$. The proof uses the fact that D_N *normalizes* $\mathcal{L}$, not membership of D_N in $\mathcal{C}_3(K)$.

6 Stable tameness and the scope of the conclusion

Proposition 6.1. *For each constructed rectifier,*

$$\Phi_* \times \mathrm{id}_{\mathbb{A}^3} \in \mathrm{Tame}_6(K), \qquad * = T, H, N \ (N \geq 1).$$

Proof. Berson–van den Essen–Wright proved that, in characteristic zero, the subgroup of $\mathrm{Aut}_K(\mathbb{A}^3)$ generated by automorphisms fixing one coordinate is contained in $\mathrm{Tame}_6(K)$ after adjoining three variables [9, Corollary 4.9]. Each of A_0, A_N fixes x. The outer factors of our rectifiers are tame. Elementary automorphisms fix two coordinates, and the affine group is generated by elementary linear transformations, scalings of one coordinate, and translations of one coordinate; each such generator fixes at least one coordinate. Thus every Φ_* belongs to the stated subgroup. □

Three additional variables suffice; their minimal number is not determined here. Combining the proofs gives the working conclusion

$$\Phi_* \in \big(\mathcal{C}_3(K) \cap \mathrm{Aut}_K(\mathbb{A}^3)\big) \setminus \mathrm{Tame}_3(K), \qquad \Phi_* \times \mathrm{id}_{\mathbb{A}^3} \in \mathrm{Tame}_6(K).$$

These are properties of the *chosen maps*. They do not rule out the existence of another tame rectifier of the same curve.

The existence of wild automorphisms in three variables and of wild coordinates is already established in the literature. Membership of the wild Nagata automorphism in $\mathcal{C}_3$ has also been proved previously [11, Proposition 6.8]. Accordingly, these results are not claimed to solve a general tameness problem, the general stable tameness conjecture, or the problem of generators for the full group Cr_3. The content of this application is the simultaneous characterization of explicit rectifiers of specific curves, together with a proof for every index of the constructed family.

7 Reproducibility and the limits of the computational certificate

Exact arithmetic. The new code uses Python and SymPy with rational coefficients and the explicit relation $a^2 - a + 1 = 0$. No floating-point arithmetic is used. Comments, error messages, and function descriptions are in English. The checks use explicit exceptions and remain enabled when Python is run with the `-O` flag.

What is checked computationally.

1. Both congruences for P_x, F_x, exact divisions, the compact base inverse map, and substitutions into the trefoil parametrization.
2. The harmonic identity, the transferred pair, exact divisions in the forward and inverse directions, and recovery of the curve from the axis.
3. For an explicitly specified finite range of indices: the recursion for p_N, polynomiality of C_N, invertibility of the centre modulo the full power x^N, construction of the correction, certificates of polynomiality for both maps, and exact substitutions into the parametrizations.
4. The normalization of the inner coordinate in the wildness proof, the order in x of its quadratic coefficient, and the monomial parity used in the Cremona argument.
5. Negative controls: a corrupted congruence and a numerator that does not admit the required exact division must be detected, including in `-O` mode.

Scope of the computational certificate. For large maps, the code uses factored certificates: polynomiality on both sides and identities after localization. This establishes global invertibility in an integral domain; fully expanding all six component compositions into large polynomials is neither required nor claimed for every N. The assertion for all N follows from induction and the proofs in the text, not from the finite range of tests. The general theorems on wildness, stable tameness, and monomial transformations are used as published mathematical results. The accompanying programs do not formally verify those theorems in a proof assistant. An explicit word in elementary automorphisms of six variables has not been computed here: the existence of such a word follows from [9].

Running the new appendix. From the directory `appendices/2026-10-28-cremona-program`:

```
python -m pip install -r requirements.txt
python computations/run_all.py
```

The command runs both normal and optimized modes; the precise range, software versions, results, and timings are recorded under `results/`. The supplied run checks $N = 1, 2, 3$: 90 checks passed in each mode, and their mathematical results agreed. Individual checks are described in `computations/README.md`. The verifier does not require network downloads while it runs. This English edition reuses the existing computation records; its preparation is not presented as a new execution of those computations.

Reruns of the original preprint's verifiers. The following commands were run from the full repository root when Appendix 2 was prepared:

```
python -u verify.py
python -u verify_printed_formulas.py
```

Both runs completed successfully. The second reads the actual right-hand sides of the formulas in the original LaTeX source and checks all twelve components of the forward and inverse rectifiers. The logs are `results/upstream_compact.log` and `results/upstream_printed.log` in the new appendix. Other old logs are included in the full archive as historical material; their presence does not mean that every earlier search, GAP script, or Macaulay2 script was rerun during the preparation of Appendix 2.

Building the English edition and auditing the files. The English source is `latex_en/main.tex`; it is compiled with two pdfLaTeX passes. The helper `build_pdf_en.py` automates this build. The standalone Overleaf ZIP contains all sections, the bibliography, and a README. It uses standard TeX fonts, so no bundled font files are needed. The preamble follows the original preprint: `article`, 11 pt, A4, 24 mm margins, theorem environments, page headers, and clickable links. The files `PROVENANCE.md`, `audit/claim_status.md`, and `SHA256SUMS.txt` record the original source version, verification scope, and checksums. The English edition is additionally documented in `ENGLISH_EDITION.md` and `SHA256SUMS_EN.txt`. The original main text and Appendix 1 are not rewritten. This remains a separate research branch for further corrections and collaborative work.

References

[1] S. S. Abhyankar, T. T. Moh. *Embeddings of the line in the plane*. J. Reine Angew. Math. **276** (1975), 148–166. doi:10.1515/crll.1975.276.148.

[2] M. Suzuki. *Propriétés topologiques des polynômes de deux variables complexes, et automorphismes algébriques de l'espace* $\mathbb{C}^2$. J. Math. Soc. Japan **26** (1974), 241–257. doi:10.2969/jmsj/02620241.

[3] A. R. Shastri. *Polynomial representations of knots*. Tohoku Math. J. (2) **44** (1992), 11–17. doi:10.2748/tmj/1178227371; open-access PDF. §1; §3, Example 1.

[4] S. M. Bhatwadekar, A. K. Dutta. *On residual variables and stably polynomial algebras*. Comm. Algebra **21** (1993), 635–645. doi:10.1080/00927879308824585. Theorem 3.2; connection with the existence of a companion coordinate.

[5] E. Edo, S. Vénéreau. *Length 2 variables of $A[x, y]$ and transfer*. Ann. Polon. Math. **76** (2001), 67–76. doi:10.4064/ap76-1-6; open-access PDF. Theorems 1–4, Proposition 2, §5.

[6] A. van den Essen, P. van Rossum. *Triangular derivations related to problems on affine n-space*. Proc. Amer. Math. Soc. **130** (2002), 1311–1322. doi:10.1090/S0002-9939-01-06212-8. Theorem 4.1, Corollary 4.2, Conjectures 5.1–5.3.

[7] U. U. Umirbaev, J.-T. Yu. *The strong Nagata conjecture*. Proc. Natl. Acad. Sci. USA **101** (2004), 4352–4355. doi:10.1073/pnas.0308157101. Main Theorem; Theorems 5–6.

[8] M. Mella, E. Polastri. *Equivalent birational embeddings*. Bull. London Math. Soc. **41** (2009), 89–93. doi:10.1112/blms/bdn107; arXiv:0906.4858. Theorem 1 and Corollary 1.

[9] J. Berson, A. van den Essen, D. Wright. *Stable tameness of two-dimensional polynomial automorphisms over a regular ring*. Adv. Math. **230** (2012), 2176–2197. doi:10.1016/j.aim.2012.04.017; arXiv:0707.3151. Corollary 4.9.

[10] E. Edo. *Coordinates of $R[x, y]$: Constructions and Classifications*. Comm. Algebra **41** (2013), 4694–4710. doi:10.1080/00927872.2012.700981; arXiv:1206.5876. Proposition 2, Theorem 8(1) and proof of Theorem 8(3).

[11] J. Blanc, I. Hedén. *The group of Cremona transformations generated by linear maps and the standard involution*. Ann. Inst. Fourier **65** (2015), 2641–2680. doi:10.5802/aif.2999; open-access PDF. Theorem 1.1; Propositions 6.5, 6.8.

[12] D. Lewis. *A unified approach to embeddings of a line in 3-space*. Comm. Algebra **52** (2024), 189–200. doi:10.1080/00927872.2023.2239344; arXiv:2203.08178 (2022 preprint). Lemma 1, Theorems 2, 12; introduction for prior results.

[13] S. Oda. *Note on the candidate counter-example in the cancellation problem for affine spaces posed by Arno Van den Essen*. arXiv:1201.4198. Withdrawn by the author on 9 May 2020; not used as a proof.

[14] I. Dorofeev. *Polynomial rectification of the trefoil and a harmonic figure-eight*. Research repository, 2026. Pinned source version; Appendix 1. Project material; not an externally peer-reviewed publication.